Control Variates: Example 1

Monte Carlo Statistical Methods

Sai Sampath Kedari
sampath@umich.edu

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1 Example 1: Estimating an Integral Using Control Variates

Consider estimating the integral

$$I_g = \int_0^1 g(x)\,dx, \qquad g(x) = e^{-x^2}.$$

There is no closed-form expression for this integral using elementary functions. However, we can reinterpret it as an expectation with respect to a uniform random variable.

Let

$$X \sim \mathcal{U}[0,1],$$

then

$$I_g = \mathbb{E}[g(X)] = \mathbb{E}[e^{-X^2}].$$

Under this formulation, a natural Monte Carlo estimator for I_g is

$$\hat{I}_g = \frac{1}{n}\sum_{i=1}^{n} g(X_i), \qquad X_i \overset{iid}{\sim} \mathcal{U}[0,1].$$

The function $g(x)$ is smooth and decreasing on $[0,1]$, and its numerical plot is shown below. For reference, the exact value is computed using sympy

1.1 Exact Integral via Symbolic Computation

We can compute the exact value of the integral using `sympy`:

```
x = sympy.Symbol('x')
exact = sympy.integrate(sympy.exp(-x**2), (x, 0, 1))
print("Exact Integral = ", exact)
print("Approximate Value = ", float(exact))
```

```
Exact Integral =  sqrt(pi)*erf(1)/2
Approximate Value =  0.746824132812427
```

1.2 Control Variate Choice and Its Exact Expectation

To improve the Monte Carlo estimate of I_g, we introduce a control variate that is easy to integrate analytically. A natural choice, motivated by the first–order Taylor expansion of e^{-x^2}, is

$$h(x) = 1 - x^2.$$

Since $X \sim \mathcal{U}[0,1]$, the expectation of $h(X)$ is

$$\mathbb{E}[h(X)] = \int_0^1 (1 - x^2)\,dx.$$

Evaluating this integral,

$$\mathbb{E}[h(X)] = \left[x - \frac{x^3}{3}\right]_0^1 = 1 - \frac{1}{3} = \frac{2}{3}.$$

Thus the control variate $h(x)$ has a known closed-form expectation,

$$I_h = \mathbb{E}[h(X)] = \frac{2}{3}.$$

1.3 Comparison of the Target Function and Control Variate

Let us compare the behavior of the target function $g(x) = e^{-x^2}$ and the control variate $h(x) = 1-x^2$ on the interval $[0, 1]$. Both functions are smooth, decreasing, and concave on this domain, which visually explains why they tend to be strongly correlated. This similarity is the key reason why $h(x)$ is an effective control variate for estimating I_g.

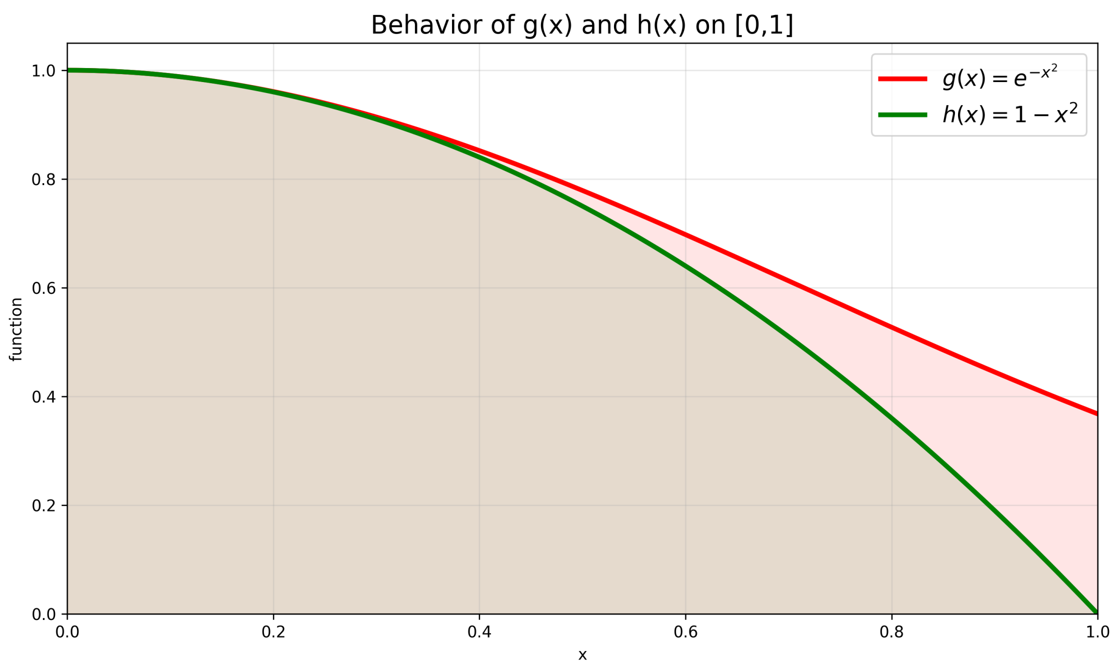

Figure 1: Behavior of $g(x) = e^{-x^2}$ and $h(x) = 1 - x^2$ on $[0, 1]$. Their similar shapes indicate strong correlation, which leads to effective variance reduction using control variates.

1.4 Monte Carlo Baseline Estimate

Before applying control variates, we begin with a standard Monte Carlo estimate of

$$I_g = \mathbb{E}[g(X)], \qquad g(x) = e^{-x^2}, \quad X \sim \mathcal{U}[0, 1].$$

Using $n = 100$ independent samples, the Monte Carlo estimator and its standard error are computed below:

```
num_samples = 100
mc_g = monte_carlo(num_samples, np.random.rand, g_evaluator=g)

# estimated Monte Carlo standard error
MC_stderr = np.std(mc_g.evaluations) / np.sqrt(num_samples)
```

```python
# error relative to the exact value
MC_Estimate_err = np.abs(float(exact) - mc_g.estimate)

print(f"Estimate = {mc_g.estimate:6f}")
print(f"Exact Error = {MC_Estimate_err:6f}")
print(f"MC Estimated Error = {MC_stderr:6f}")
```

```
Estimate = 0.755224
Exact Error = 0.008400
MC Estimated Error = 0.019401
```

This baseline will allow us to compare the variance reduction achieved by the control variate method in the following subsection.

1.5 Control Variate Estimate

We now apply the control variate method using

$$h(x) = 1 - x^2,$$

whose expectation $\mathbb{E}[h(X)] = 2/3$ is known exactly. The optimal control variate weight α^* is estimated from the sample covariance between $g(X)$ and $h(X)$, using the same Monte Carlo samples.

The control variate estimator is

$$\hat{I}_{CV} = \hat{I}_g + \alpha^* \left(\hat{I}_h - \mathbb{E}[h(X)] \right),$$

which has variance

$$\mathrm{Var}(\hat{I}_{CV}) = \frac{1}{n} \mathrm{Var}(g)\,(1 - \rho^2),$$

where ρ is the sample correlation between $g(X)$ and $h(X)$.

The Python code used to compute the control variate estimate is shown below:

```python
num_samples = 100
cv_g = control_variate(num_samples,np.random.rand,g,h,
    h_expected=h_expected_val)

# Estimated standard error of the control variate estimator
cv_stderr = np.sqrt(cv_g.cov[0, 0] / num_samples * (1 - cv_g.correlation**2))

# Error relative to the exact value
cv_Estimate_err = np.abs(float(exact) - cv_g.mc_g.estimate)

print(f"CV Estimate = {cv_g.mc_g.estimate:6f}")
print(f"Exact Error = {cv_Estimate_err:6f}")
print(f"CV Estimated Error = {cv_stderr:6f}")
```

```
CV Estimate = 0.736836
Exact Error = 0.009988
CV Estimated Error = 0.002603
```

Using $n = 100$ samples, the Monte Carlo estimator has a standard error of approximately 2.0×10^{-2}, while the control variate estimator reduces this to about 2.6×10^{-3} an order-of-magnitude improvement using the same number of samples.

1.6 Variance Comparison of Monte Carlo and Control Variates

To visualize the effect of variance reduction, we generate 500 independent Monte Carlo and Control Variate estimates using $n = 100$ samples each. The resulting histograms are shown in Figure 2.

The reduction in variability is dramatic. Quantitatively, we observe

$$\sigma_{\mathrm{MC}} \approx 2.01 \times 10^{-2}, \qquad \sigma_{\mathrm{CV}} \approx 2.39 \times 10^{-3}.$$

This corresponds to roughly an **eight- to ten-fold reduction** in the standard deviation of the estimator. The CV estimates form a narrow, concentrated cluster around the exact value, while the MC estimates remain widely spread.

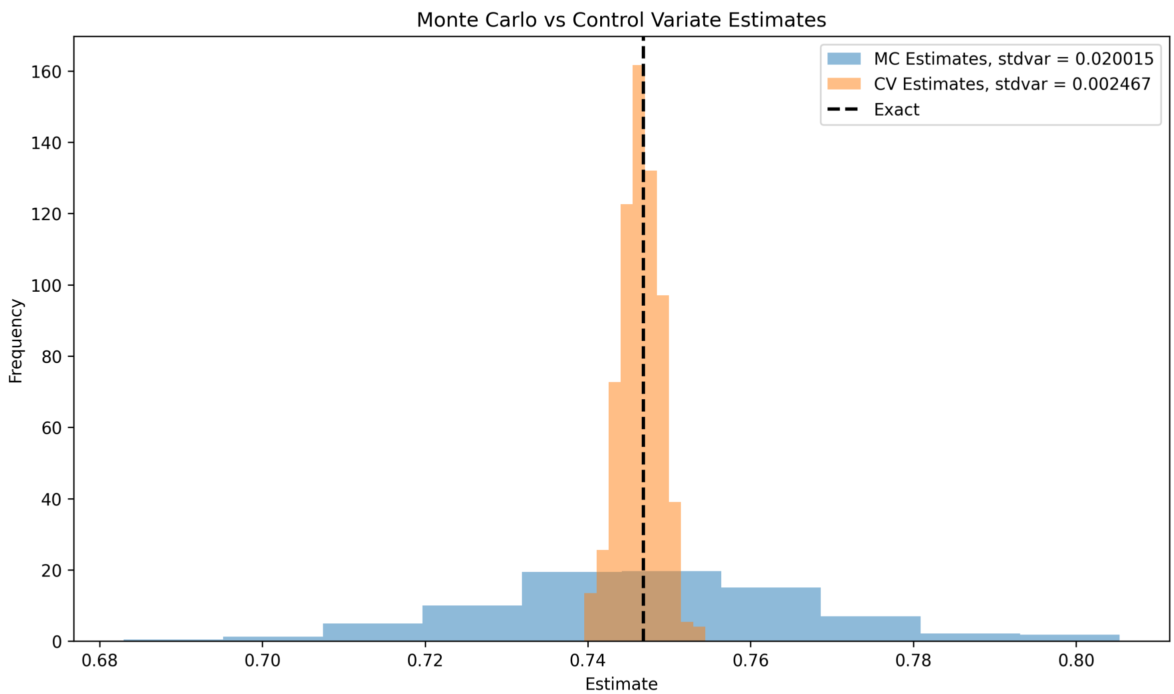

Figure 2: Comparison of 500 Monte Carlo and Control Variate estimates for $I_g = \mathbb{E}[e^{-X^2}]$ with $n = 100$ samples per trial. Control variates significantly tighten the distribution of estimates.